Mikolaj Hilgert

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Skills

Development & ML	Python, C# (.NET), Java (Spring), TypeScript, Scala, Apache Spark, Apache Airflow, LangChain, Pinecone, Postgres, MongoDB, PyTorch, scikit-learn
Cloud & Infrastructure:	Microsoft Azure, Google Cloud, Docker, Kubernetes, Git, Terraform, DevOps, MLOps, Databricks, SQL (MySQL)
Languages	English (native), Polish (native), Czech (proficient), German (B1)

Experience

Graduate Data Science Intern (Full-Time) — **Nov 2025 — Present**
ASML — *Veldhoven, The Netherlands*

- Completing a Master's thesis internship focused on the predictive maintenance of high-tech lithography equipment.
- Conducted comprehensive data analyses to answer business questions related to upgrade effectiveness and power recovery trends in chip production.
- Utilised various machine learning methods including deep learning for time series forecasting of machine power degradation, successfully outperforming existing estimation methods.

Software Developer (Part-Time) — **Sep 2024 — Dec 2025**
CY2 — *s-Hertogenbosch, The Netherlands*

- Managed the development of a Microsoft Teams–based student app built primarily on Microsoft Azure, unifying LMS, SIS, and other systems into a single interface for students and staff.
- Developed a multi-tenant, LLM-powered chatbot platform that empowers educational organizations to create tailored chatbots using their own documents and datastores.

Data Science Intern (Full-Time) — **Feb 2024 — Jul 2024**
ASML — *Veldhoven, The Netherlands*

- Thesis title: Impact of parallel unscheduled machine downs on the eXtreme long down (XLD) frequency of DUV lithography systems.
- Conducted data analysis and developed an algorithm to quantify the impact of staffing on XLD frequency.
- Researched predictive maintenance methods and implemented machine learning techniques to predict XLD occurrences.

Trainee Software Engineer (Part-Time) — **Mar 2023 — Nov 2023**
CM.com (prev. Building Blocks) — *Tilburg, The Netherlands*

- Worked on improving the AI Decisioning Engine, which is used to deliver mass consumer data personalisations, increase conversion rates, order value, and lower returns for clients like Coca-Cola, Corendon, Blokker, and Basic-Fit.
- Enhanced the IntelliJ IDEA development plugin, which was utilised on a daily basis by over 20 data scientists, expediting the development cycle and streamlining the delivery of tailored solutions for all Building Blocks' clients.

Software Engineering Intern (Full-Time) — **Sep 2022 — Feb 2023**
CM.com (prev. Building Blocks) — *Tilburg, The Netherlands*

- Implemented an advanced user-detail encryption module utilising Apache Spark within an existing production codebase, improving data security and GDPR compliance.
- Developed a data loader for Building Blocks' proprietary platform for files stored in Azure Blob Storage.

Notable projects

Agentic Voice Agent Secretary — **Sep 2025 — Jan 2026**
University project

- Spearheaded the development of a real-time patient intake and triage platform, streamlining the appointment booking process for Dutch medical clinics.
- Designed and implemented the underlying AI voice agent, utilising LangChain to bridge OpenAI's response generation with Twilio and ElevenLabs for seamless voice interactions.

Full-Stack Dashboard with ETL Data Pipeline Processing — **Sep 2021 — Jan 2022**
Group University project

- Collaborated in a high-performing team of 5 students following the Scrum Agile framework.
- Contributed to the design and development of the ETL data pipeline using Python and Apache Airflow.

Education

Master of Science in Data Science in Business and Entrepreneurship, *JADS (Tu/e & Tilburg University)* — Aug 2024 — Jul 2026
Pre-master in Data Science in Business and Entrepreneurship, *JADS (Tu/e & Tilburg University)* **Grade: 8.0** — Jan 2023 — Jun 2023
Bachelor of Science in ICT & Software Engineering, *Fontys University of Applied Science* **Grade: 9.2** — Sep 2020 — Jul 2024